

San Diego Weather Helper for Photographers Proposal

Developer: Mimi Do

Target Audience: San Diego photographers and clients booking shoots

Technology Stack: Python, Microsoft Power BI, PowerQuery, DAX, Open-Meteo API

Background

San Diego, California, is world-renowned for its scenic coastal landscapes, dramatic cliffs, and vibrant urban backdrops, making it a prominent hub for professional and hobbyist photography. However, photography is fundamentally an art form dictated by light and atmospheric conditions.

While standard weather applications provide basic temperature and precipitation forecasts, they fail to deliver the domain-specific insights photographers require. Elements like cloud cover drastically alter light quality, while localized humidity and wind can ruin intricate hair and makeup styling before a shoot even begins.

Building an automated data pipeline to extract, transform, and visualize atmospheric data specifically for creative professionals optimizes the session booking workflow. This project ensures photographers can confidently select optimal shooting windows, advise clients on wardrobe and styling prep, and eliminate costly, weather-induced reshoots.

Business Problem

San Diego is a beautiful city with copious landscapes that attract photographers. Weather and lighting issues can cause delays, poor photo quality, and unhappy clients. This application will help photographers better plan their shoots and allow clients to adjust to the weather beforehand.

Objectives

- Create an intuitive web-based weather application using Microsoft PowerBI
- Integrate real-time weather data from Open-Meteo API for San Diego, CA
- Develop photoshoot recommendation algorithms based on weather parameters
- Provide users with lighting intel throughout the day
- Warn users of unfavorable humidity and precipitation
- Recommend hair and makeup styles for varying weather conditions

Methodology / Technical Approach

Power BI Dashboard

- **Executive KPI Cards:** Card visuals showing immediate temperature, real-time humidity, and precipitation percentages with conditional icon alerts
- **Interactive Visual Timeline:** A dual-axis line and clustered column chart mapping temperature trends and hourly precipitation probabilities over the forecast timeline
- **Slicers & Filters:** Interactive page slicers allowing photographers to filter by *Desired Shoot Type* (e.g., Beach Portrait, Indoor Studio, Golden Hour Landscape) or *Weather Tolerance* to isolate the exact days and hours optimal for booking

Recommendation Engine Logic (DAX-Powered)

The Power BI semantic model will use rule-based **DAX calculated columns and measures** to generate real-time recommendations:

- **Lighting Quality Diagnostics (DAX Measure):**
Cloud Cover] < 20 && [Is Golden Hour] = 1, "Golden Hour (Peak Light)",
IF([Cloud Cover] >= 40 && [Cloud Cover] <= 80 && [Precipitation] = 0, "Soft Diffused Portrait Light",
IF([Cloud Cover] < 15 && [Is Mid Day] = 1, "Harsh Highlights (Diffusers Required)", "Standard Daylight"))
- **Shoot Type & Location Suitability:**
 - *Beach/Coastal Shoots:* Suitable only if Wind Speed < 12 mph and precipitation is 0%.
 - *Indoor Studio Shoots:* Recommended if precipitation > 60% OR temperature > 85°F.
- **Hair & Makeup Styling Alerts (DAX Calculated Column):**

- Supports multiple codes per description (comma-separated expansion)
- Enables human-readable weather condition display

1. Implementation Plan

Week 1: Foundation & API Ingestion

- Review Open-Meteo documentation and finalize San Diego coordination variables.
- Write the core Python extraction script using the requests library.
- Establish successful ingestion and initial flattening of JSON responses via pandas.

Week 2: Core Engineering & Storage

- Set up local data storage architectures (SQLite database or clean CSV outputs).
- Construct the localized translation logic functions to process numeric weather code references.
- Connect the local storage directory directly to Power BI via the Power Query editor.

Week 3: Power BI Data Modeling & DAX Engine

- Design the internal data model schema inside Power BI Desktop.
- Program the custom photography recommendation rules using calculated DAX columns.
- Build foundational aggregate DAX measures for tracking average humidity, peak winds, and light quality percentages.

Week 4: Power BI Layout & Interactive Visuals

- Design the interactive time-series charts, cross-filtering slicers, and KPI blocks.
- Set up visual conditional formatting patterns to flag hair, makeup, and equipment weather risks.
- Test report interactivity, cross-filtering responsiveness, and clean up visual asset alignment.

Week 5: Final Evaluation & Submission

- Finalize end-to-end data pipeline validation from raw API call to Power BI report refresh.
- Assemble full technical pipeline project documentation.
- Record the final video presentation demonstration walk-through and submit all deliverables.

Expected Outcomes

- **Data Pipeline:** Python script that automates data fetching, flattening, and staging routines
- **Structured Data Model:** Power BI dataset with cleanly mapped lookup keys and optimal data types
- **Interactive Power BI Report:** A.pbix dashboard utilizing advanced cross-filtering, custom slicers, and dynamic DAX alerts explicitly tailored for San Diego photography planning
- **Documentation Suite:** A comprehensive engineering report detailing implementation parameters alongside an instructional video demo.

Success Metrics

Technical Metrics

- **API Reliability:** >99% successful API calls
- **Page Load Time:** <3 seconds initial load
- **Data Accuracy:** Weather data within 15 minutes of current time

User Experience Metrics

- **User Engagement:** Average session duration >5 minutes
- **Recommendation Acceptance:** >70% of users interact with activity suggestions
- **Return Usage:** >50% user retention rate
- **User Satisfaction:** >4.5/5 star rating

Conclusion

This weather helper bridges the gap between raw data engineering and creative application by transforming complex meteorological metrics into practical business intelligence for San Diego photographers. By substituting complex custom JavaScript or web app deployments with an agile, high-performance Microsoft Power BI architecture, this project delivers an efficient tool that minimizes session scheduling uncertainty, keeps clients happy, and helps local creative businesses run smoothly.

Next Steps

1. **Stakeholder Review:** Present proposal to project sponsors and stakeholders
2. **Technical Feasibility Assessment:** Verify API capabilities and library compatibility
3. **Prototype Development:** Create minimum viable product (MVP) with core features
4. **User Research:** Gather feedback on activity preferences and UI requirements
5. **Project Kickoff:** Begin Week 1 development with clear milestones and deliverables